FLUXORA - COMPLETE END-TO-END ENGINEERING SUMMARY

Futuristic Financial Flow Tracking, Roommate Split Ledger & AI Financial Advisor

1. Project Identity, Platform Configuration & Architecture

- Application Name & ID: Fluxora (applicationId: com.aistudio.financetrack.qwrxnv, namespace: com.example) synchronized across strings.xml, settings.gradle.kts, and metadata.json.
- Tech Stack: 100% Kotlin and Jetpack Compose with Material Design 3 (M3), Coroutines/StateFlow, DataStore Preferences, WorkManager, Jetpack Credential Manager, Biometric API, and Firebase (Auth, Firestore, AI Logic, App Check).
- Adaptive Launcher Icon: Custom vector foreground and background adaptive launcher icons configured in mipmap-anydpi-v26.
- Edge-to-Edge & Responsive UI: Configured enableEdgeToEdge() in MainActivity with proper WindowInsets handling and responsive layouts.

2. Futuristic Visual Design System & 3D Interactive Components

- 8 Signature Themes (AppTheme): Midnight Emerald (Default), Cyber Violet, Solar Gold, Arctic Cyan, Crimson Pulse, Obsidian Rose, Aurora Borealis, and Monochrome Slate, with full Dark and Light mode support in Theme.kt.
- FinancialHero3DScene: Custom interactive 3D Cash-Flow Reactor / Gyroscopic Orbital Core rendered on Canvas with touch-drag 3D rotation, animated particle streams, income/expense orbital rings, and live net balance telemetry.
- DynamicIsland: Floating top status capsule surfacing live cash-flow health, active budget warnings, and quick navigation.
- Indian Currency Formatting: FinancialEngine.formatIndianCurrency formats minor units (paise) into standard Indian numbering grouping (e.g., INR 39,050.00) with optional sign prefixes.

3. Authentication, Phone Verification & Security Architecture

- Multi-Modal Firebase Auth (AuthManager): Supports Google Sign-In via Jetpack Credential Manager (GetSignInWithGoogleOption), Email/Password authentication, and Firebase Phone Auth OTP verification.
- Per-User Phone Verification & Ownership Scoping: PreferencesManager scopes verified E.164 phone numbers per user UID (isPhoneVerifiedFlow, getVerifiedPhoneFlow, getPhoneOwnerFlow) and prevents duplicate phone number claiming across different accounts.
- International Phone Validation (PhoneValidation.kt & CountryPhoneInputField.kt): Supports country code search and selection (defaulting to India +91), national digit extraction, and strict E.164 normalization.
- Biometric Lock & Inactivity Monitor (BiometricAuthManager, BiometricAuthScreen, AppLockMonitor): Hardware biometric/device credential lock gate with configurable inactivity timeout and automatic state cleanup on sign-out.
- Sensitive Amount Masking (LocalHideSensitiveAmounts): Global CompositionLocal privacy toggle masking monetary figures across all screens.
- Firebase App Check (AppCheckConfiguration): Play Integrity provider for release builds and Debug provider for development builds.

4. Cloud Data Layer Migration & Financial Domain Engine

- Room-to-Firestore Data Layer Consolidation: Removed legacy Room SQLite database/DAOs (AppDatabase, TransactionDao, CategoryDao, BudgetDao,

SavingsGoalDao, RecurringBillDao, RoommateDao, HouseholdMemberDao, NotificationDao) and commented out unused Room dependencies in app/build.gradle.kts.

- Unified FinanceRepository: Reactive StateFlow + FirebaseFirestore persistence layer with zero-trust firestore.rules enforcing authenticated ownership and strict schema validation.
- Transactions & Date Validation: Supports INCOME and EXPENSE records in minor units (1..MAX_AMOUNT_MINOR). DateValidator.validateTransactionDate allows past/today dates and rejects future-dated transactions with clear user feedback.
- Categories: 12 built-in default categories (Housing, Food & Dining, Groceries, Transportation, Education, Shopping, Entertainment, Health, Bills & Utilities, Travel, Subscriptions, Other) plus custom user categories.
- Budgets & Savings Goals: Monthly category budgets with 80% warning and 100% exceeded alerts; savings goals with automatic 25%, 50%, 75%, and 100% milestone tracking.
- Atomic Recurring Bills: Weekly, Monthly, and Yearly recurring bills. markBillAsPaid atomically records the expense transaction, advances nextDueDate, and deduplicates rapid repeat invocations within a 10-second window.

5. Roommate Split Ledger, Connection Requests & UPI Payments

- Shared Expense Splitting: Supports EQUAL, EXACT, PERCENTAGE, and SHARES split methods. Enforces strict validation that participant split amounts sum exactly to totalAmount.
- Bidirectional Roommate Connection Requests (RoommateConnectionRequest): Enforces strict sender/recipient query separation (isIncomingFor vs. isOutgoingFor), blocks self-requests by phone or UID, blocks duplicate pending requests, blocks unverified phone numbers, prevents senders or third parties from accepting/declining requests, protects against remote sync tampering (syncRemoteConnectionRequest), and automatically adds accepted roommates to HouseholdMember.
- UPI Payment Handoff (PaymentManager): Validates UPI VPAs (alex@okhdfcbank, etc.) and amount bounds, launching standard Android upi://pay deep-link intents for instant settlement.

6. AI Financial Advisor (Firebase AI Logic & Gemini 2.5 Flash)

- Daily AI Saving Insight: Uses Firebase.ai.generativeModel("gemini-2.5-flash") with 30s timeout, automatic retry on transient errors, and friendly fallback messages for App Check, quota, or network issues.
- Interactive Gemini Chat Assistant (GeminiChatDialog & GeminiChatViewModel): Conversational financial assistant aware of user budgets, bills, and spending patterns, with clean manual state reset (resetUiState).

7. Notification System, CSV Data Portability & Legal/Creator Screens

- Granular Notification Engine (NotificationManager, SystemNotificationHelper, BillReminderWorker, BillActionReceiver): Master notification switch preserves individual category toggles (Budget, Bill, Savings, Roommate, Insight) when disabled and re-enabled, suppresses disabled categories, and deduplicates alerts by relatedId.
- CSV Export & Formula Injection Defense (exportToCsv): Exports transactions to CSV while prefixing leading =, +, -, and @ characters with a single quote to neutralize spreadsheet formula injection.

- CSV Import & Batch Deduplication (importFromCsv): Validates columns and amounts, reports malformed rows, and silently skips duplicate rows within the same import batch.
- Legal Compliance & Contact Routing (LegalScreens.kt): 8-section Privacy Policy and Terms & Conditions with accordion UI, zero drafting placeholders, and official mailto: intent routing (openContactEmail) for Ayush Mishra and Sunaina Bansal.
- Authors & Credits (AuthorsScreen.kt): Showcases creators Ayush Mishra, Sunaina Bansal, and Rohan Kumar (with LinkedIn integration).

8. Verification, Compilation & 74-Test Robolectric Suite

- Full Project Compilation: Verified clean debug APK compilation via compileDebugKotlin and assembleDebug.
- Automated Test Suite (ExampleRobolectricTest & SecurityAndPaymentTest): 74 unit and Robolectric tests validating themes, roommate connection security, recurring bill idempotency, CSV import/export safety, UPI validation, notification deduplication, phone verification scoping, and legal compliance.